

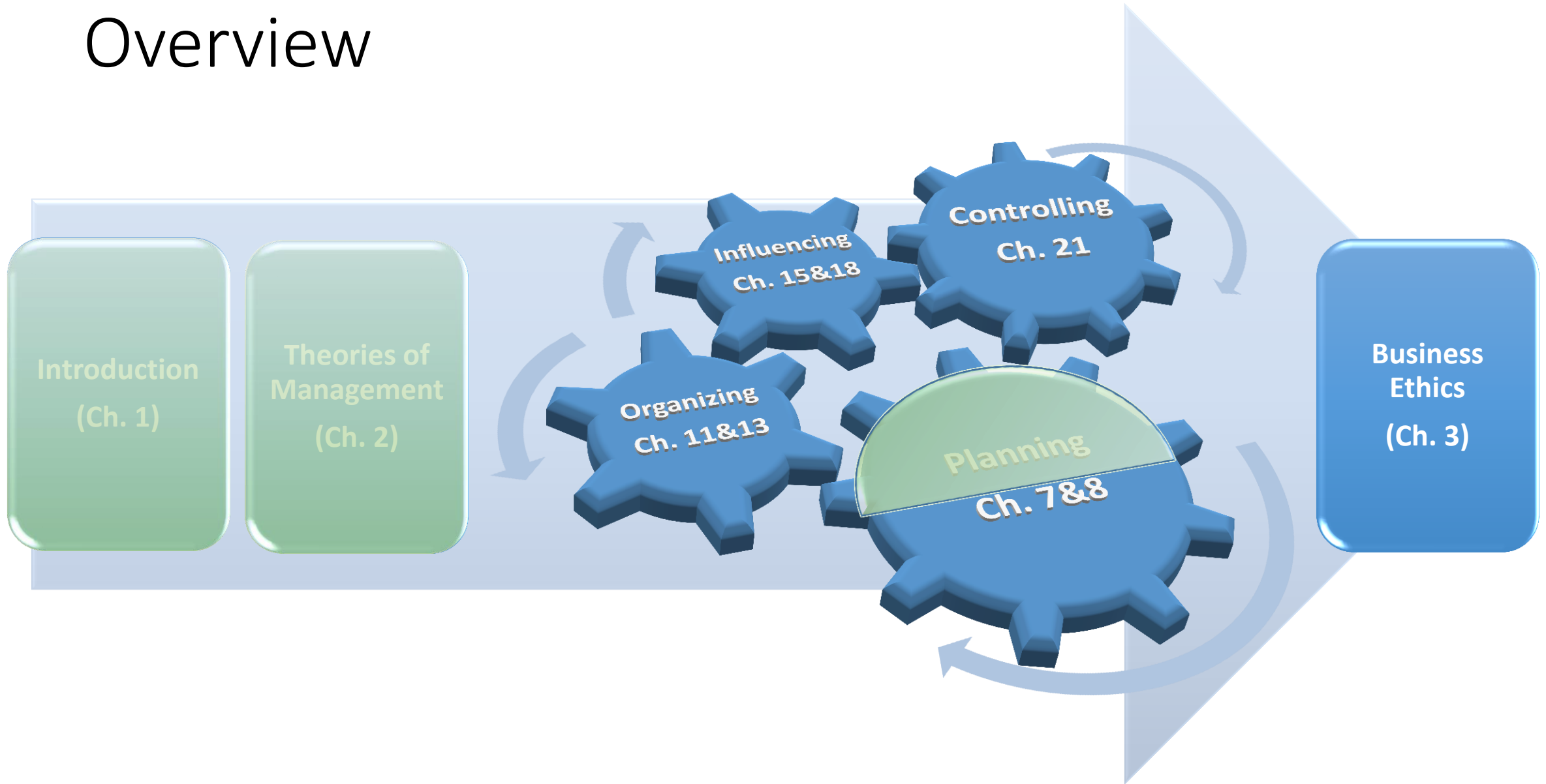
Chapter 8

Making Decisions

By

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Overview



Objectives

1. *A fundamental understanding of the term decision*
2. *An understanding of each element of the decision situation*
3. *An ability to use the decision-making process*

Definitions

- **Decision:**

Is a choice made between two or more available alternatives.

- **Decision Making:**

Is the process of choosing the best alternative for reaching objectives.

Decisions Types

- **Decision can be categorized based on:**
 - How much time a manager must spend making them?
 - What proportion of the organization must be involved in making them?
 - Which organizational functions they focus on?
- **Programmed decision:**

routine and repetitive
- **Non-programmed decision:**

one-shot decision

Decisions Types

Types of Decisions	Decision-Making Techniques	
	Traditional	Modern
Programmed: Routine, repetitive decisions Organization develops specific processes for handling them	1. Habit 2. Clerical routine: Standard operating procedures 3. Organization structure: Common expectations A system of subgoals Well-defined information channels	1. Operations research: Mathematical analysis models Computer simulation 2. Electronic data processing
Nonprogrammed: One-shot, ill-structured, novel policy decisions Handled by general problem solving processes	1. Judgment, intuition, and creativity 2. Rules of thumb 3. Selection and training of executives	1. Heuristic problem-solving techniques applied to: Training human decision makers Constructing heuristic computer programs

The Responsibility for Making Organizational Decisions

Who will make the decision?

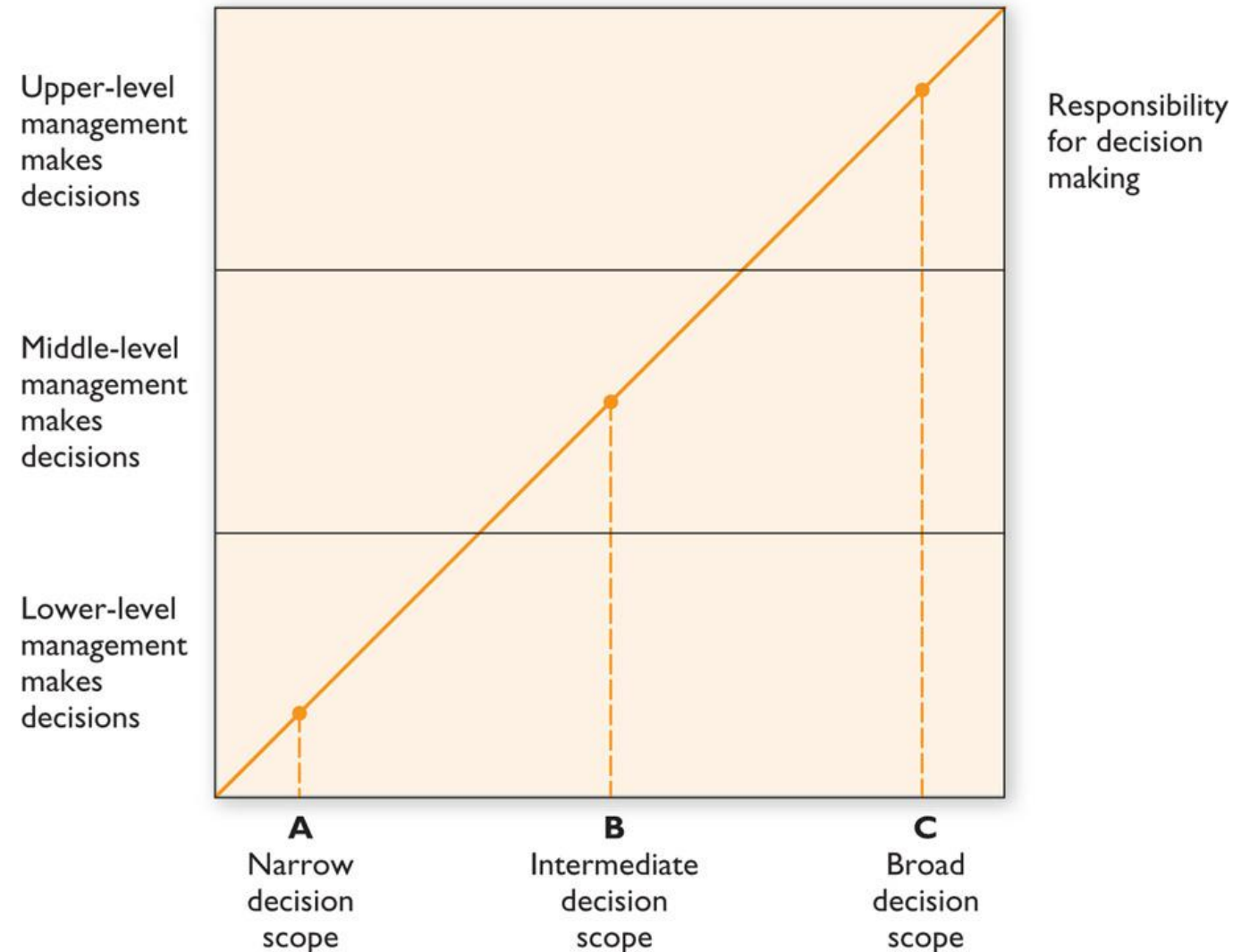
1. Decision Scope:

The proportion of the total management system that the decision will affect.

2. levels of management:

Lower, Middle, or Upper Level.

Making Decision Responsibility Rationale



Consensus

- **Definition:**

Is an agreement on a decision by all the individuals involved in making that decision.

- **Advantages:**

- It focuses “several heads” on the decision.
- Employees are more likely to be committed to implementing a decision if they helped make it.

- **Disadvantages:**

- it often involves time-consuming discussions relating to the decision

Elements of the Decision Situation

- The Decision Makers

Weak DM usually is receptive, exploitative, hoarding, or marketing oriented

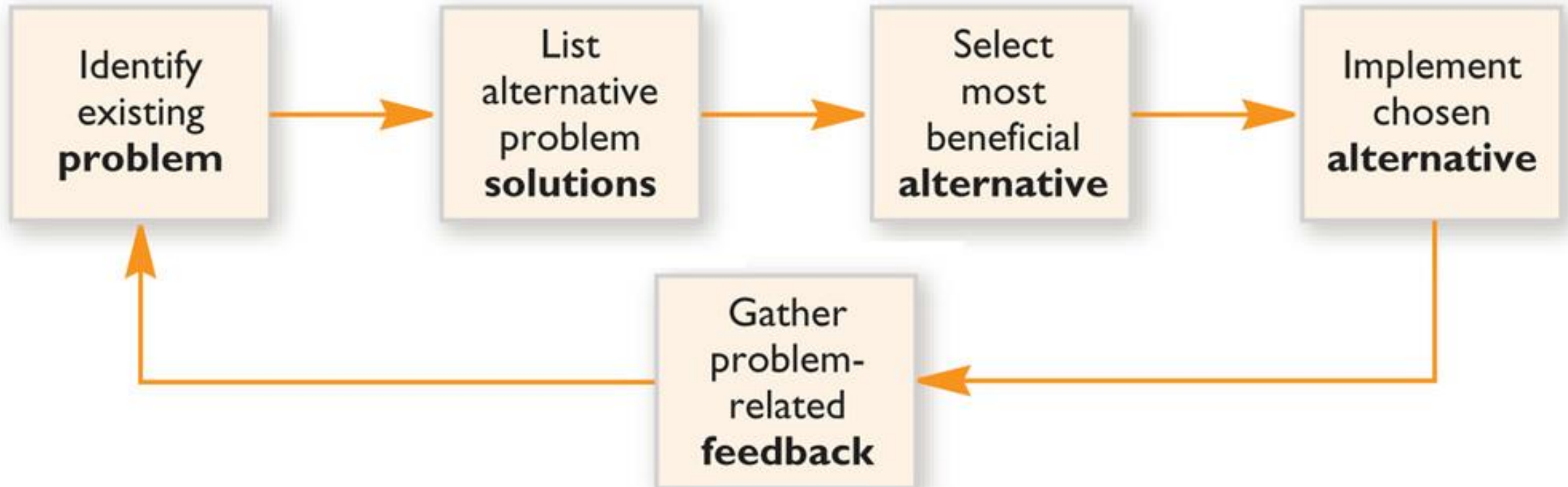
- Goals to Be Served

- Relevant Alternatives

- Ordering of Alternatives

- Choice of Alternatives

The Rational Decision-Making Process Model



The Rational Decision-Making Process

Assumptions

1. Humans are economic beings with the objective of maximizing satisfaction or return.
2. All alternatives and their possible consequences are known.
3. Decision makers have some priority system to guide them in ranking the desirability of each alternative.

The Rational Decision-Making Process

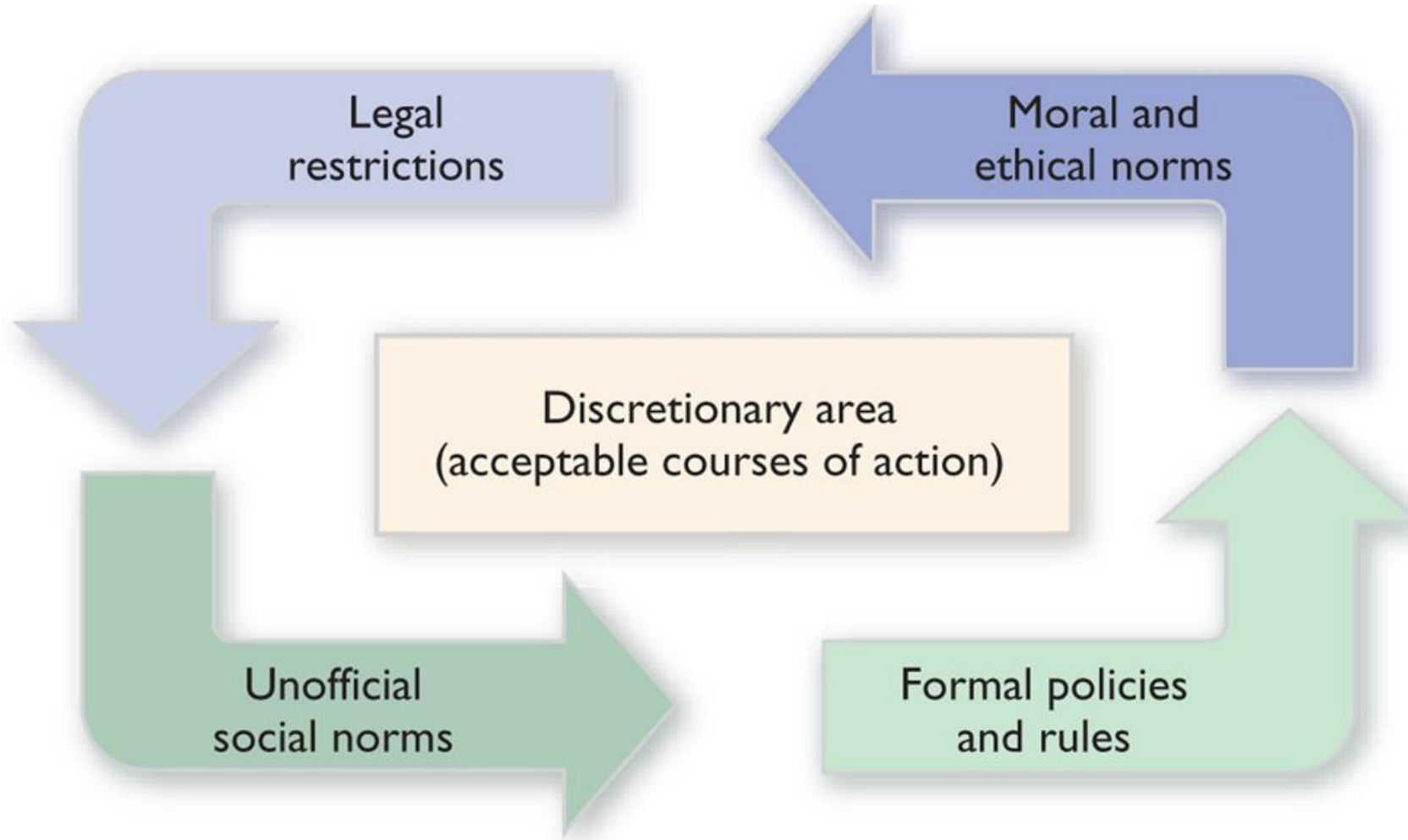
Listing Alternative Solutions

Limitations on the number of problem-solving alternatives available:

1. Authority factors.
2. Biological or human factors.
3. Physical factors.
4. Technological factors.
5. Economic factors.

The Rational Decision-Making Process

Listing Alternative Solutions



The Rational Decision-Making Process

Listing Alternative Solutions

Which is the better, to have higher or fewer number of alternatives?

- Generally, generating more alternatives would usually lead to more effective decision making.
- However, having too many alternatives may actually demotivate decision makers.

The Rational Decision-Making Process

Selecting the Most Beneficial Alternative

Alternatives should be evaluated in these steps:

- List the potential effects of each alternative.
- Assign a probability factor to each of the potential effects.
- Compare each alternative's expected effects and the respective probabilities of those effects.

The Rational Decision-Making Process

Implementing the Chosen Alternative

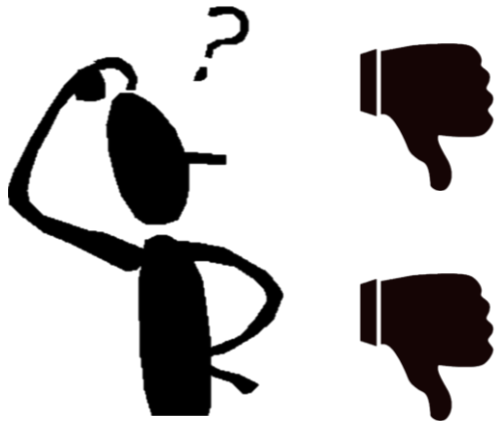
Develop an action plan to implement the chosen alternative, giving it all supportive chances of success.

The Rational Decision-Making Process

Gathering Problem-Related Feedback

Feedback determines the implemented alternative's effect on problem
If problem is not solved, seek out and implement another alternative

DM Conditions: Risk and Uncertainty

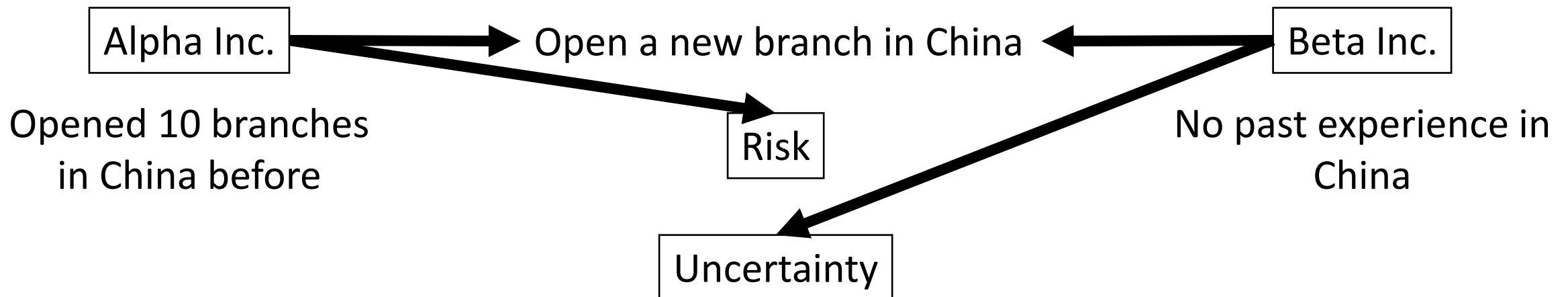


- **Risk:**

Refers to situations in which statistical probabilities can be attributed to alternative potential outcomes.

- **Uncertainty:**

Refers to situations where the probability that a particular outcome will occur is not known in advance.



Decision-Making Tools

- **Subjective:**

Intuition
Experience
Rule of Thumbs

- **Objective:**

Probability Theory
Decision Trees

Decision-Making Tools

Probability Theory

- Is a decision-making tool used in risk situations.
- Probability refers to the likelihood that an **event** or **outcome** will actually occur.

$$EV = I * P$$

- *EV*: is the expected value for the alternative considered.
- *I*: is the income that alternative would produce.
- *P*: probability of producing that income

Decision-Making Tools

Probability Theory

Alternative (locations)	Potential income	Probability of income	Expected value of alternatives
A	\$90,000	.2	\$18,000
B	75,000	.4	30,000
C	60,000	.8	48,000
$I \quad \times \quad P \quad = \quad EV$			

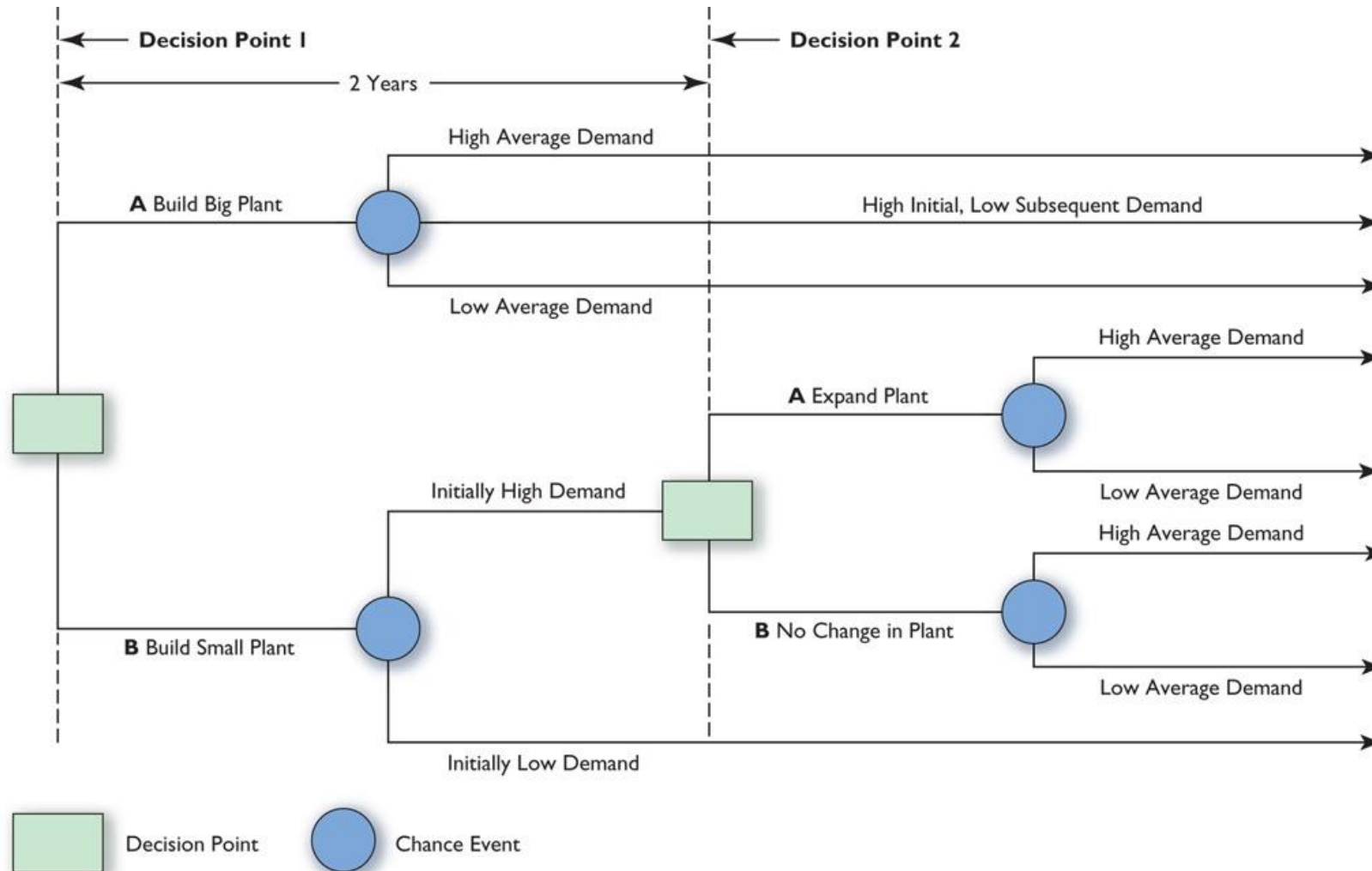
Decision-Making Tools

Decision Trees

- Is a graphic decision-making tool typically used to evaluate decisions involving a series of steps.
- Stygian Chemical Company decides whether to build a small or a large plant to manufacture a new product with an expected life of 10 years.
- If the choice is to build a large plant, the company could face high or low average product demand, or high initial and then low demand.
- If the choice is to build a small plant, the company could face either initially high or initially low product demand. If the small plant is built and product demand is high during an initial two-year period, management could then choose whether to expand the plant. Whether the decision is made to expand or not to expand, management could then face either high or low product demand.

Decision-Making Tools

Decision Trees



Decision-Making Tools

Decision Trees

- The financial consequence of each different course of action must be compared.
 1. Study estimates of investment amounts necessary for building a large plant, for building a small plant, and for expanding a small plant.
 2. Weigh the probabilities of facing different product demand levels for various decision alternatives.
 3. Consider projected income yields for each decision alternative.
- ***Net expected gain***: is defined in this situation as the expected value of an alternative minus the investment cost.

Group Decision Making

Advantages and Disadvantages

- **Advantages:**

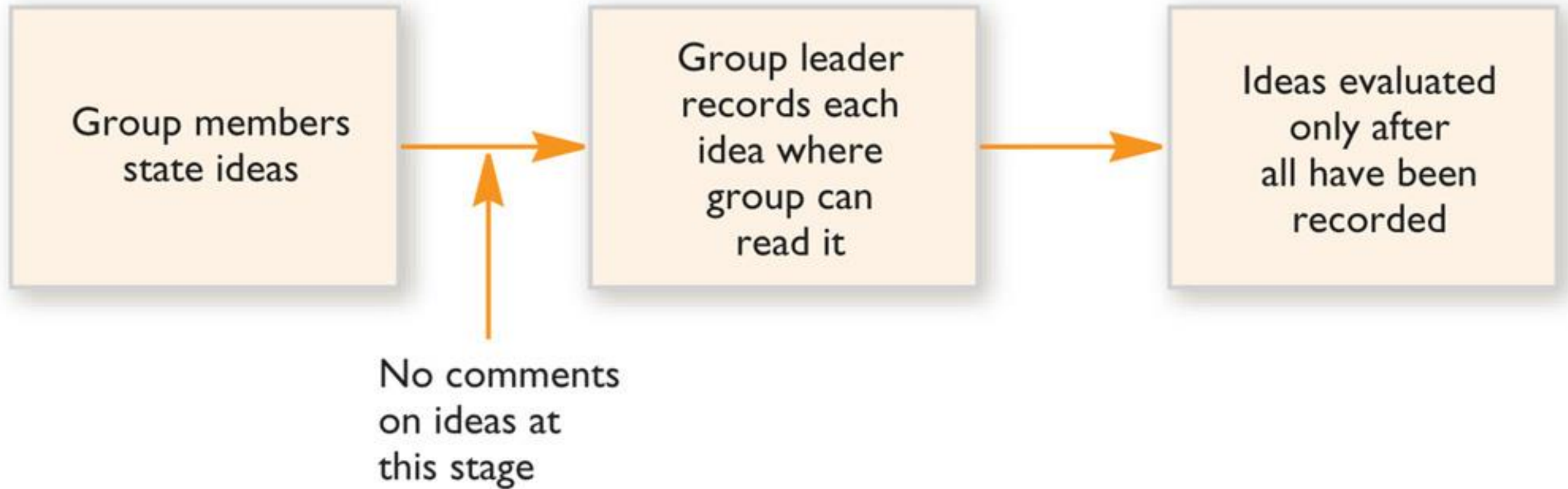
- A group can generally come up with more and better decision alternatives.
- A group can draw on collective, diverse organizational experiences as the foundation for decision making.
- The members of that group tend to support the implementation of the decision more fervently.

- **Disadvantages:**

- It takes longer to make a group decision because groups must take the time to present and discuss all the members' views.
- group decisions cost the organization more than individual decisions do simply because they take up the time of more people in the organization.
- **Groupthink:** group members' efforts to maintain friendly relationships among themselves.

Group Decision Making

Brainstorming



Group Decision Making

Brainstorming

- **Advantages:**

Encouraging the expression of as many useful ideas as possible

- **Disadvantages:**

Wasting the group's time on ideas that are wildly impractical

Group Decision Making

Nominal Group Technique

1. Each group member writes down individual ideas on the decision or problem being discussed.
2. Each member presents individual ideas orally. The ideas are usually written on a board for all other members to see and refer to.
3. After all members present their ideas, the entire group discusses these ideas simultaneously. Discussion tends to be unstructured and spontaneous.
4. When discussion is completed, a secret ballot is taken to allow members to support their favorite ideas without fear. The idea receiving the most votes is adopted and implemented.

Group Decision Making

Nominal Group Technique

- **Advantages:**

Its secret ballot, offers a structure in which individuals can support or reject an idea without fear of recrimination

- **Disadvantages:**

Group members have no way of knowing why individuals voted the way they did

Group Decision Making

Delphi Technique

1. A problem is identified.
2. Group members are asked to offer solutions to the problem by providing anonymous responses to a carefully designed questionnaire.
3. Responses of all group members are compiled and sent out to all group members.
4. Individual group members are asked to generate a new individual solution to the problem after they have studied the individual responses of all other group members compiled in Step 3.
5. Steps 3 and 4 are repeated until a consensus problem solution is reached.

Group Decision Making

Delphi Technique

- **Advantages:**

Ideas can be gathered from group members who are too geographically separated or busy to meet face to face

- **Disadvantages:**

Members are unable to ask questions of one another

Overview

